# Computer Vision with Embedded Machine Learning

## Object Detection Performance Metrics

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## Confusion Matrix

| Actual Label \ Predicted Label | Ball | Dog | Toy |
|---|---|---|---|
| Ball | 205 | 10 | 1 |
| Dog | 6 | 199 | 0 |
| Toy | 9 | 17 | 223 |

**Precision, Positive Predictive Value (PPV):** What proportion of positive predictions was actually correct?

$$Precision = \frac{TP}{TP + FP} = \frac{\square}{\square + \square}$$

$$= \frac{199}{199 + 27} = 0.881$$

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## Confusion Matrix

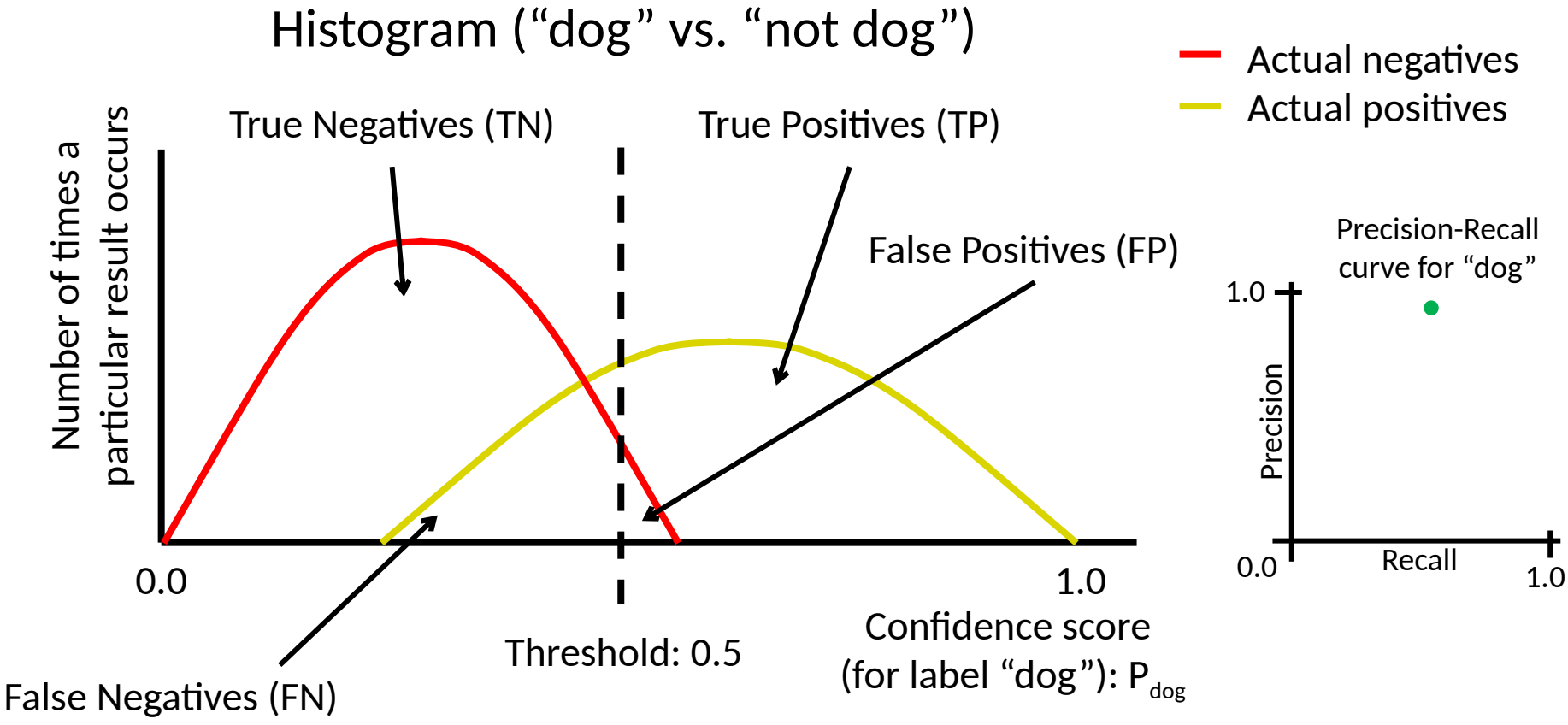
| Actual Label \ Predicted Label | Ball | Dog | Toy |
|---|---|---|---|
| Ball | 205 | 10 | 1 |
| Dog | 6 | 199 | 0 |
| Toy | 9 | 17 | 223 |

**Recall, True Positive Rate (TPR):**
What proportion of actual positives was identified correctly?

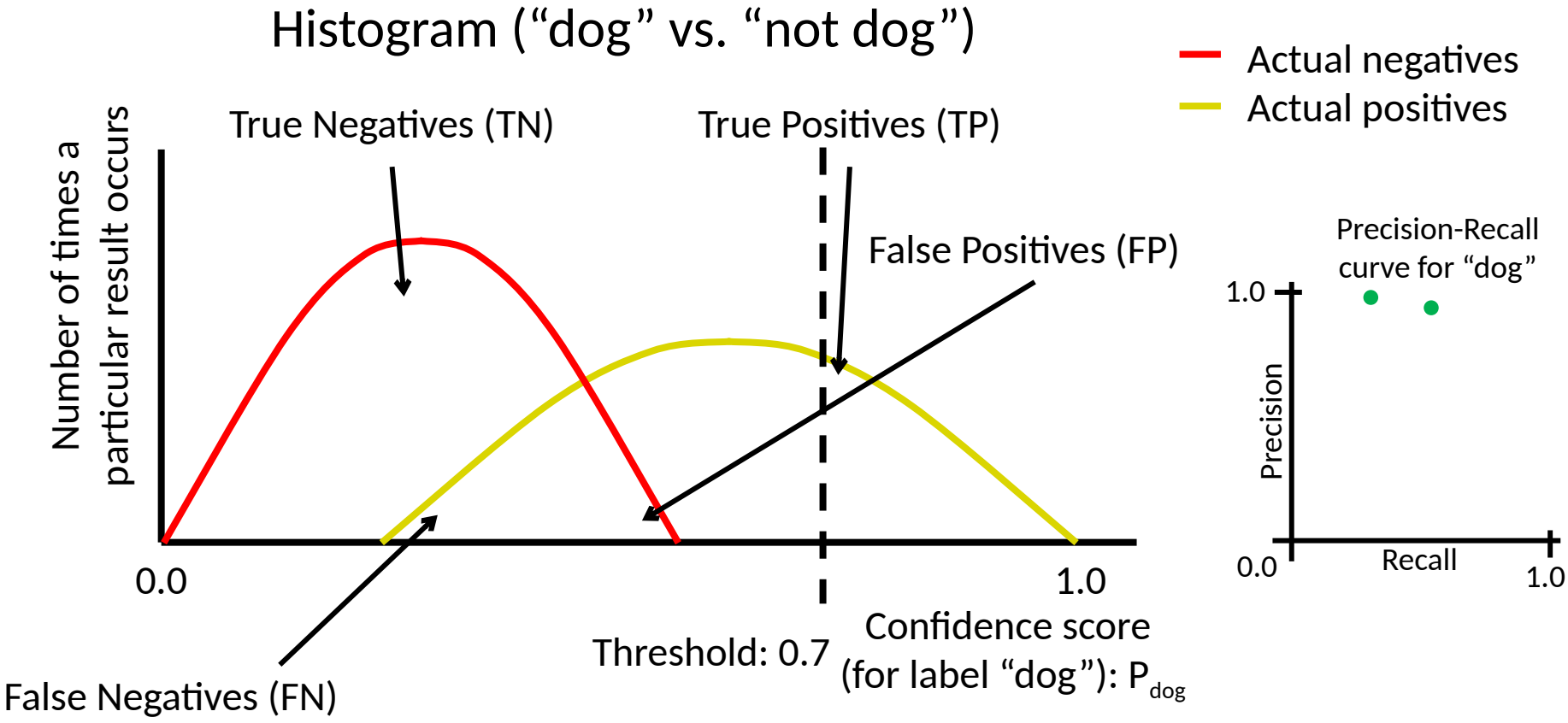
$$Recall = \frac{TP}{TP + FN} = \frac{\square}{\square + \square}$$

$$= \frac{199}{199 + 6} = 0.971$$

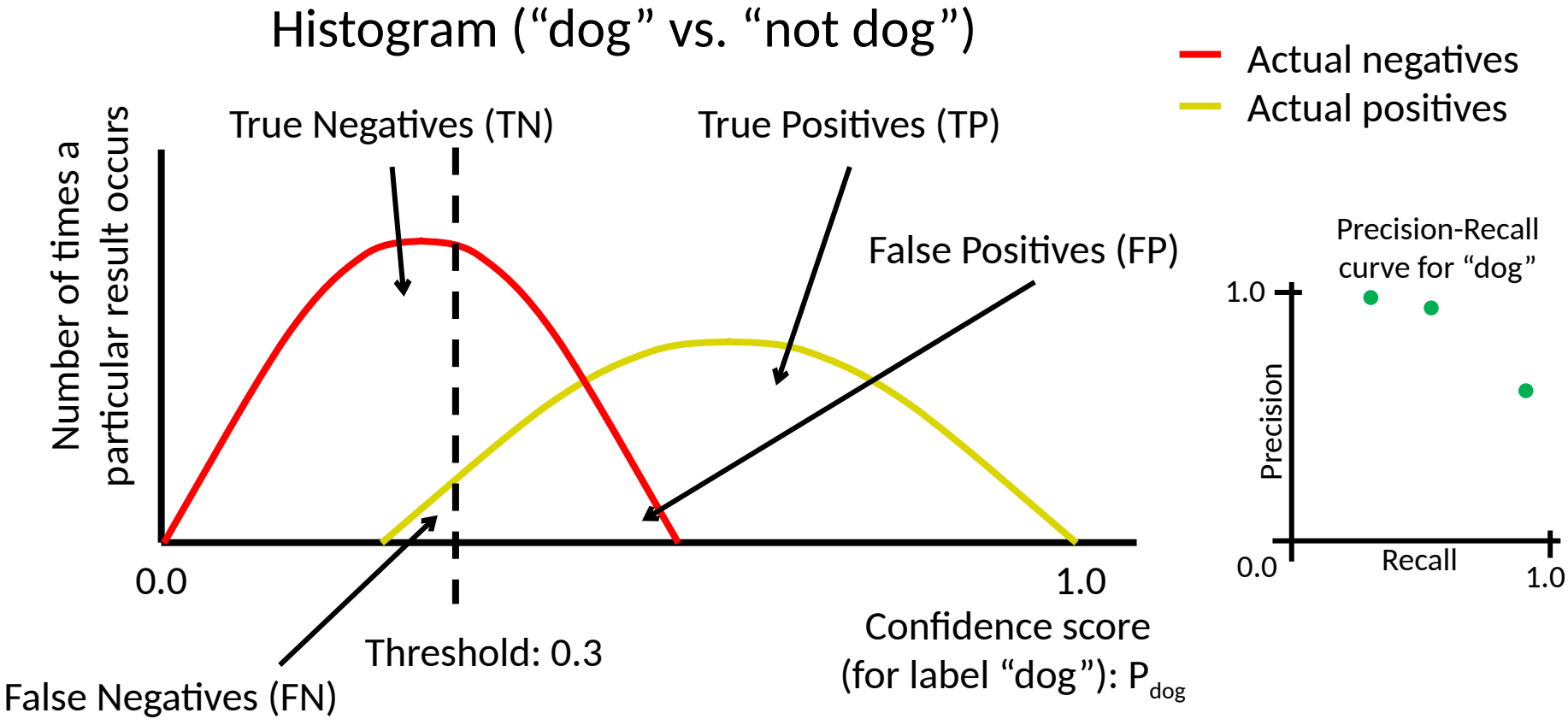
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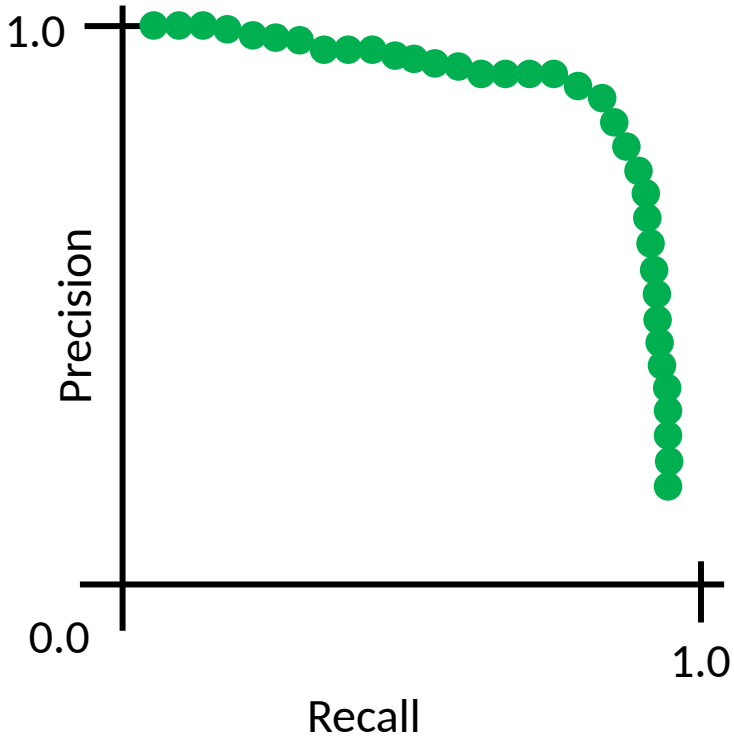



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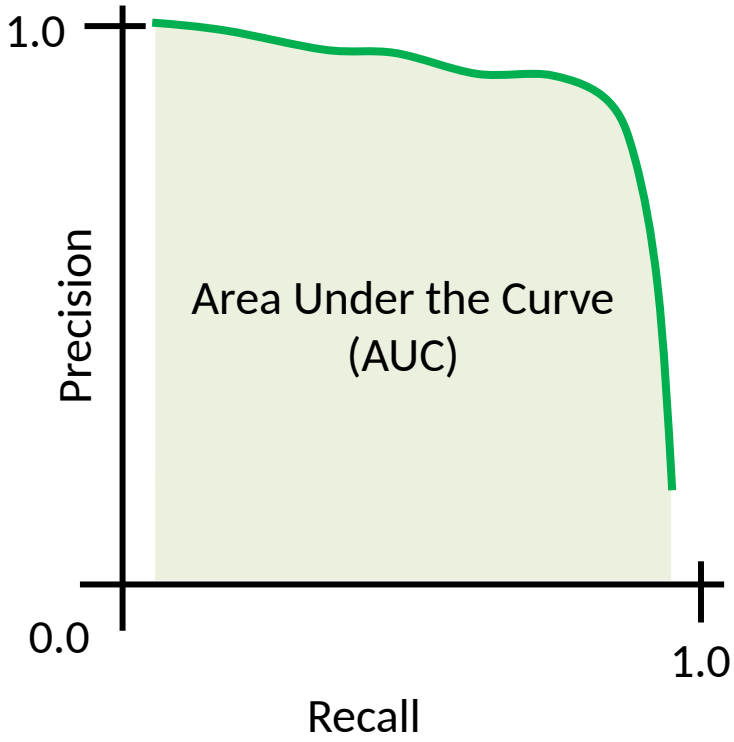

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# Precision-Recall curve for “dog”




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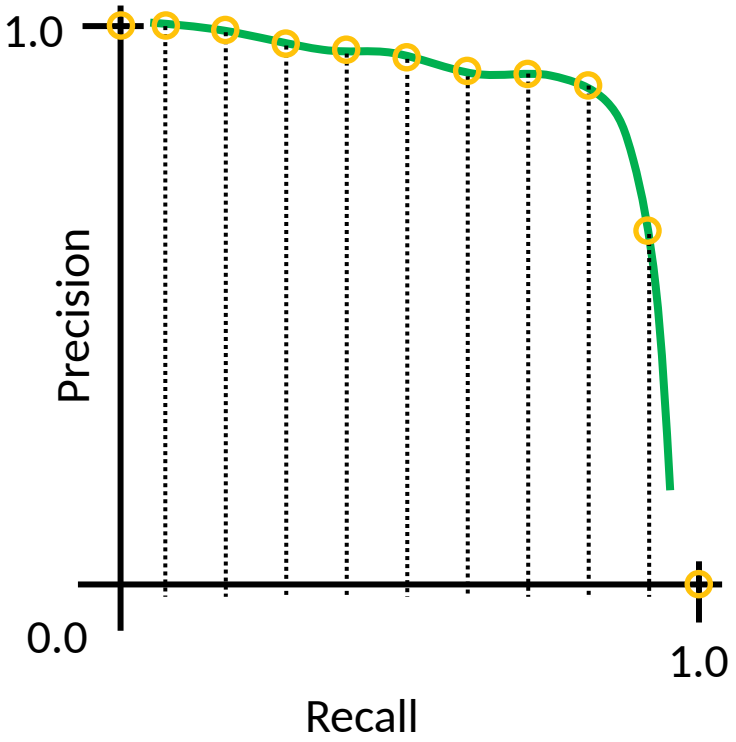
# Average Precision (AP)




$$AP = \int_0^1 p(r)dr$$

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# Average Precision (AP) approximation



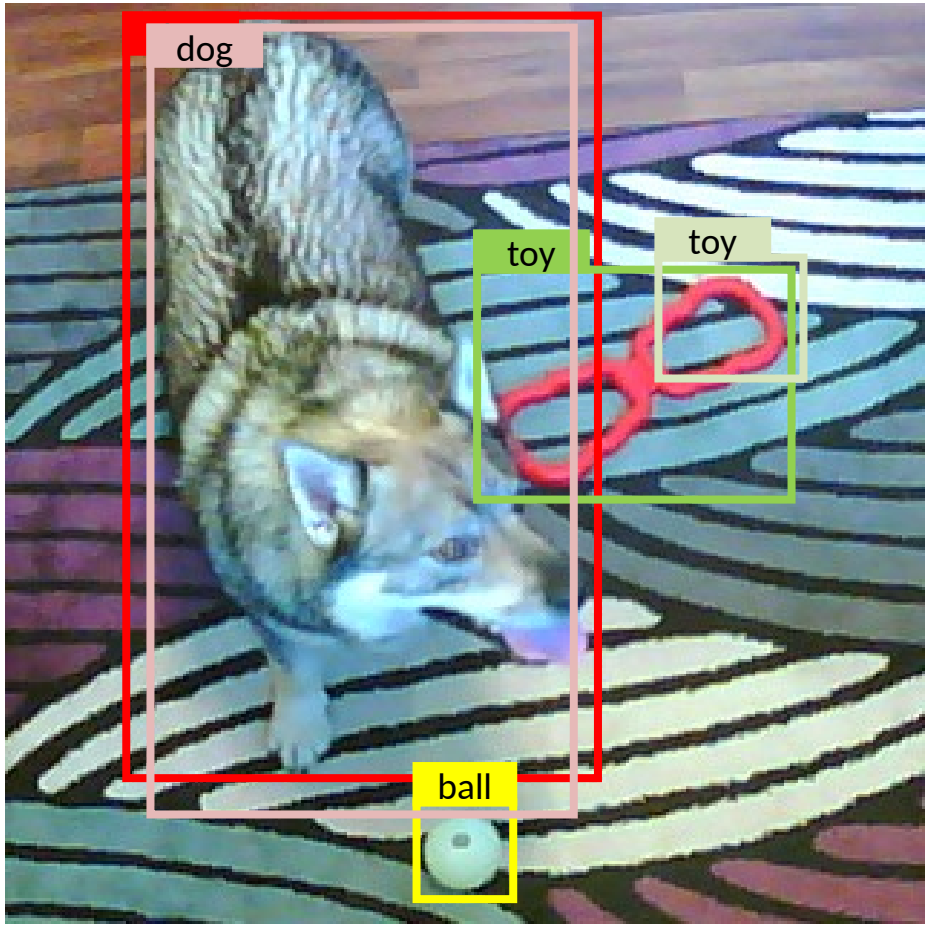

PASCAL Visual Object Classes (VOC) Challenge (2009):

$$AP = \frac{1}{11} \sum_{r_i} p(r_i)$$

where $r_i$ = [0.0, 0.1, 0.2, ..., 1.0]

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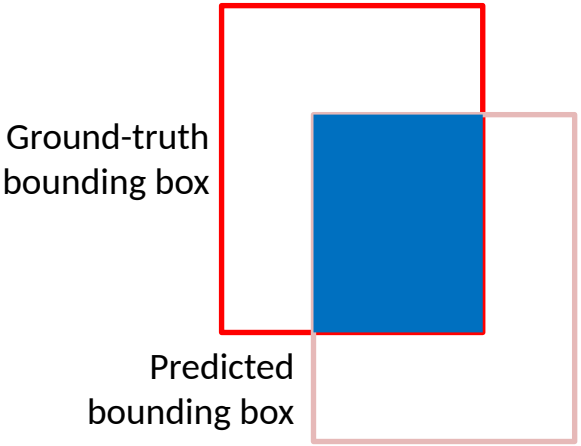
- **Object 1**
  - Class: dog
  - Bounding box
    - $(x_1, y_1)$
    - $(w_1, h_1)$
- **Object 2**
  - Class: toy
  - Bounding box
    - $(x_2, y_2)$
    - $(w_2, h_2)$
- **Object 3**
  - Class: ball
  - Bounding box
    - $(x_3, y_3)$
    - $(w_3, h_3)$



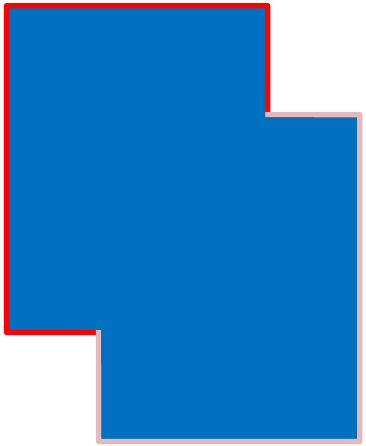

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# Intersection over Union (IoU)

**Intersection**: Area of overlap




**Union**: Area encompassed by both boxes



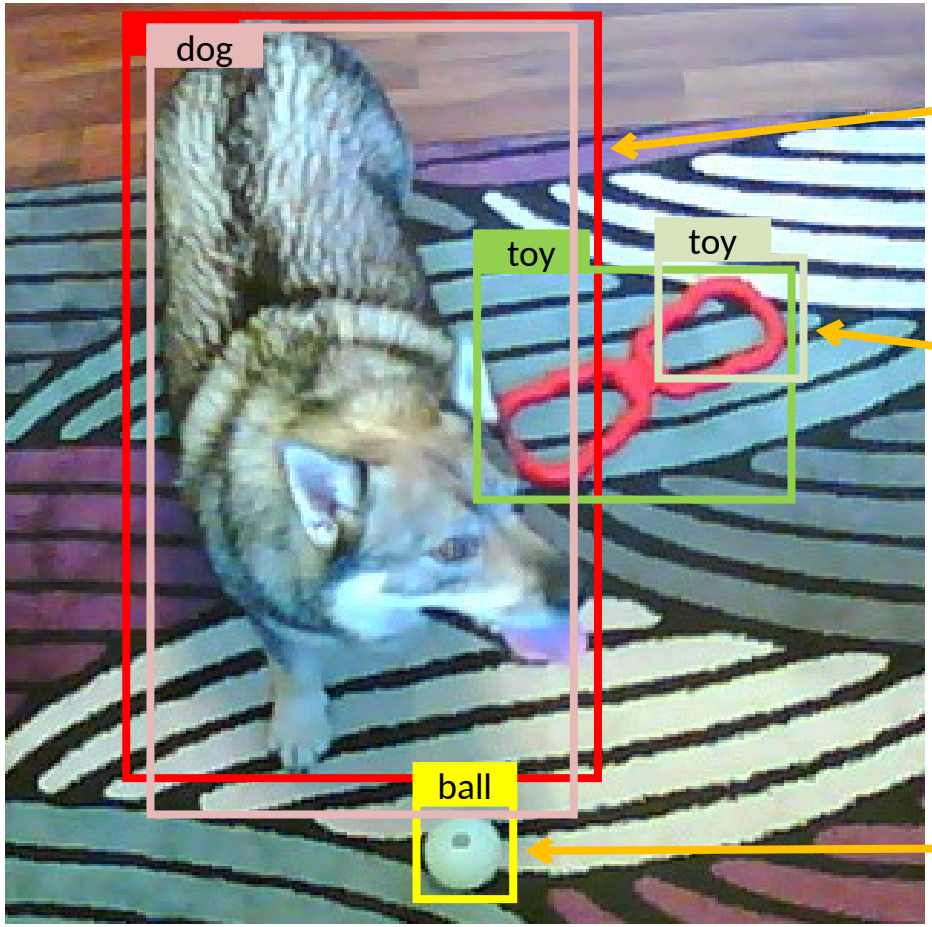
$$IoU = \frac{Intersection}{Union} = \frac{area\ of\ overlap}{area\ of\ union}$$

0.0 = complete miss
1.0 = perfect match

Object: IoU >= 0.5

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- **Object 1**
  - Class: dog
  - Bounding box
    - $(x_1, y_1)$
    - $(w_1, h_1)$
- **Object 2**
  - Class: toy
  - Bounding box
    - $(x_2, y_2)$
    - $(w_2, h_2)$
- **Object 3**
  - Class: ball
  - Bounding box
    - $(x_3, y_3)$
    - $(w_3, h_3)$




**With $Th_{IoU}$ = 0.5:**

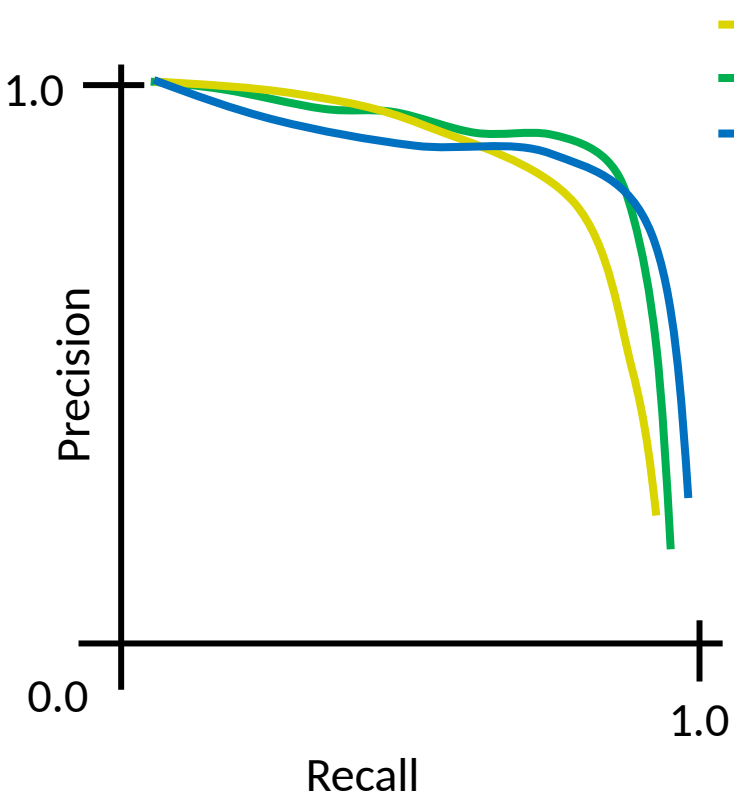
IoU = 0.9
True Positive (TP)

IoU = 0.2
False Positive (FP)

False Negative (FN)

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# Precision-Recall curve (for IoU >= 0.5)

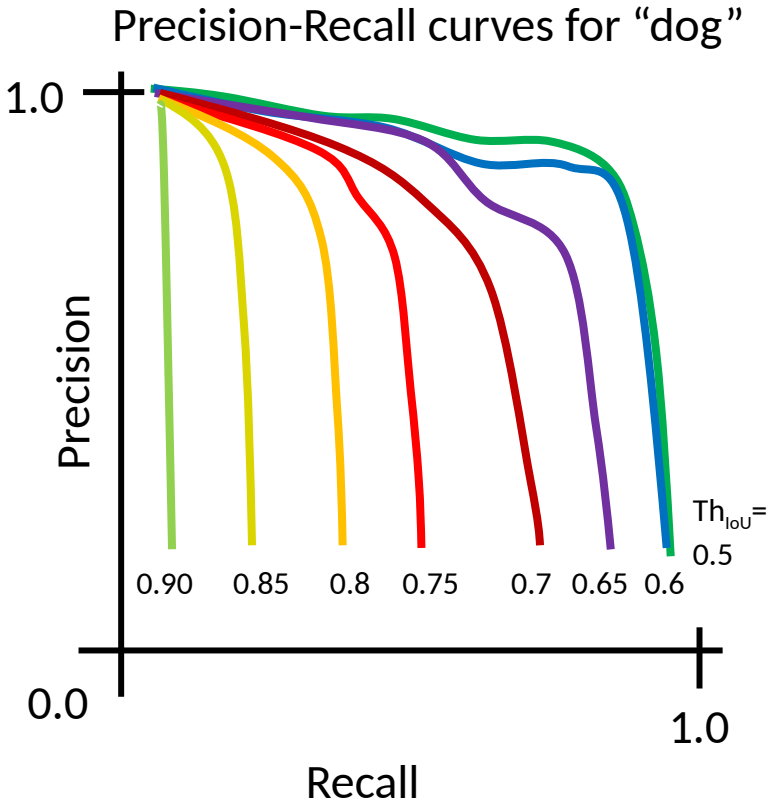



**mean Average Precision (mAP)**

$$mAP_{0.5} = \frac{1}{3}(AP_{ball} + AP_{dog} + AP_{toy})$$

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# mAP (for several IoU thresholds)




## mean Average Precision (mAP)

$$mAP_{0.5} = \frac{1}{3}(AP_{ball} + AP_{dog} + AP_{toy})$$

## COCO 2017 Challenge:

$$mAP = \frac{1}{10}\sum_{i} mAP_i$$

where $i$ = [0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95]

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